AI Systems Engineering – Innovation Driven Project Report

Title: Clinician-in-the-Loop AI Decision Support for Depression Risk
Course: AI Systems Engineering
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Project Type: Innovation Driven (INN)
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Abstract

This project presents a Clinician in the Loop AI Decision Support System designed to assist qualified mental health professionals in assessing depression risk indicators from textual clinical inputs. Rather than pursuing automated diagnosis, the system is explicitly constrained to provide probabilistic risk indicators that support but never replace clinical judgment. The work emphasizes AI Systems Engineering principles, focusing on system boundaries, human oversight, safety, fairness, and privacy in a high risk healthcare domain. A modular architecture integrating classical NLP based machine learning models within a locally deployed clinical workflow is proposed and prototyped. System level testing is conducted beyond predictive accuracy, including safety oriented evaluation, robustness analysis, and bias considerations.

The contribution of this project lies not in proposing a new predictive model, but in demonstrating how an AI component can be responsibly embedded within a safety critical clinical workflow through explicit system boundaries, human oversight, and trustworthiness mechanisms.

1. Introduction & Motivation

1.1 Background

Mental health assessment is a safety critical process in which incorrect, premature, or poorly contextualized automation can lead to severe clinical, ethical, and societal consequences. While recent advances in machine learning and natural language processing have shown promise in identifying linguistic patterns associated with depressive behavior, deploying such techniques in real clinical settings requires careful system level design.

1.2 Problem Statement

The core challenge addressed in this project is not depression detection in isolation, but the safe integration of AI generated insights into clinical workflows. Fully automated diagnosis systems pose ethical, legal, and safety risks in mental healthcare.

1.3 Project Objectives

- Design a human in the loop AI decision support system for depression risk assessment
- Preserve human agency and clinical accountability
- Ensure safety, fairness, privacy, and robustness
- Demonstrate a working proof of concept prototype

2. Stakeholders & Requirements Analysis

2.1 Stakeholders

- Mental Health Professionals: Primary users interpreting AI outputs.
- Patients: Indirect stakeholders whose textual data is analyzed.
- System Developers: Responsible for maintenance and monitoring.
- Ethical and Regulatory Bodies: Ensure compliance and ethical use.

2.2 Functional Requirements

- FR-1: The system shall accept clinician provided textual inputs associated with patient sessions.
- FR-2: The system shall generate probabilistic depression risk indicators based on textual analysis.
- FR-3: The system shall require explicit human review before any interpretation or action is taken based on AI outputs.

2.3 Non Functional Requirements

- NFR-S (Safety): The system shall not perform autonomous diagnosis or trigger automated interventions.
- NFR-F (Fairness): The system shall mitigate bias through conservative predictions and mandatory human oversight.
- NFR-P (Privacy): The system shall operate under local deployment constraints to protect sensitive patient data.
- NFR-R (Robustness): The system shall degrade gracefully under ambiguous or noisy input conditions.
- NFR-T (Transparency): The system shall present AI outputs in an interpretable and clearly contextualized manner.

3. Related Work & Existing Solutions

Existing research on AI based depression assessment largely prioritizes improving predictive accuracy using textual, acoustic, or multimodal data. However, limited attention has been paid to system level integration, human oversight, and operational risk management. This project addresses this gap by prioritizing responsible system design over isolated model performance.

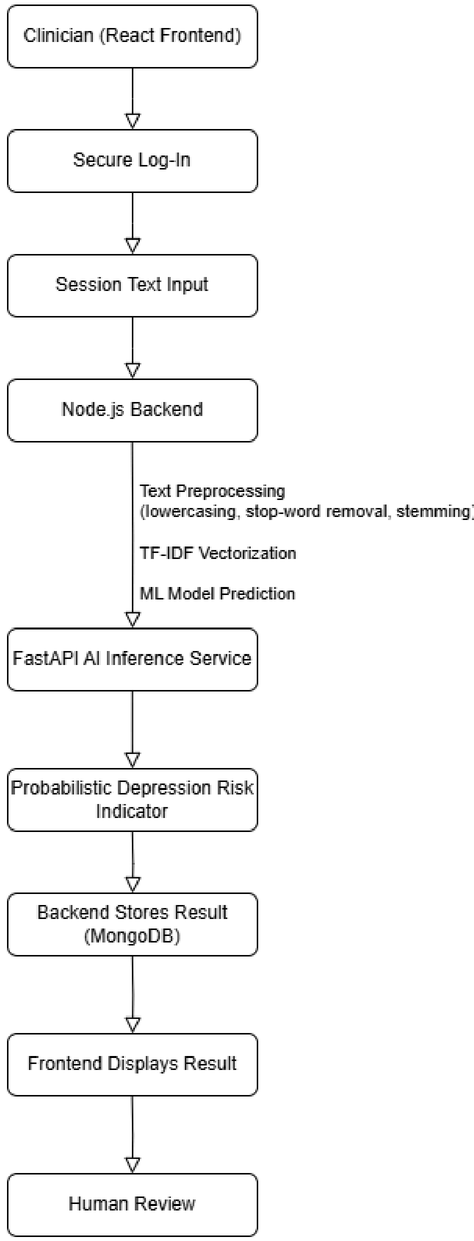
4. System Overview & Design

4.1 System Architecture

The system follows a modular, service oriented architecture:

Text Input → NLP Preprocessing → ML Model → FastAPI Inference Service → Backend Services → Frontend Interface.

Human professionals remain central in the decision loop.



4.2 Design Rationale

Classical ML models with TF IDF features were selected for interpretability, reduced hallucination risk, and deployment efficiency. Local deployment minimizes privacy risks.

4.3 System Boundary & Human Oversight

The AI component generates risk indicators only; final decisions remain under clinician control at all times.

5. AI Component Design

This section details the design and implementation of the AI component as an integral part of the overall decision support system. The focus is not only on predictive performance, but also on interpretability, safety, and deployability within a clinical workflow.

5.1 Data Sources and Labeling Strategy

The primary dataset used in this project is derived from the DAIC WOZ (Distress Analysis Interview Corpus), which contains transcribed clinical style interviews aimed at identifying psychological distress. Given the limited size of the dataset for supervised learning, additional publicly available text data related to depression and suicidal ideation was incorporated to improve linguistic diversity and model robustness.

5.2 Text Preprocessing Pipeline

To balance model effectiveness with interpretability, a classical NLP preprocessing pipeline was adopted. The preprocessing steps implemented in the notebook and deployed in the inference service include: Conversion of text to lowercase Removal of punctuation and special characters Tokenization of text into words Removal of English stop words using NLTK Stemming using the Porter Stemmer

This pipeline reduces noise while preserving clinically meaningful linguistic patterns. Importantly, the same preprocessing logic is reused consistently during both training and inference to avoid training–serving skew.

5.3 Feature Extraction

Textual inputs are transformed into numerical representations using TF IDF (Term Frequency–Inverse Document Frequency) vectorization. TF IDF was selected due to: Its transparency and ease of interpretation Lower risk of hallucination compared to generative models Compatibility with classical machine learning classifiers

The trained TF IDF vectorizer is serialized and reused at inference time to ensure consistency across the system lifecycle.

5.4 Model Selection and Training

Multiple classical machine learning models were evaluated during experimentation, including

- Naïve Bayes,
- Decision Trees,
- Random Forests,
- K Nearest Neighbors,
- Gradient Boosting,

Model selection was guided by a combination of:

- Predictive performance (accuracy, precision, recall, F1 score)
- Stability across validation splits
- Interpretability and deployment simplicity

We selected Naive Bayes model as the final one which is a trade off between performance and system level safety requirements. Rather than optimizing for maximum accuracy alone, preference was given to models whose behavior could be more easily reasoned about by system designers and clinicians.

5.5 Model Integration into the System

The trained Naive Bayes model and TF IDF vectorizer are serialized and deployed within a FastAPI based inference service. The service exposes a `/predict` endpoint that accepts textual input, applies the preprocessing pipeline, transforms the input using the stored models, and returns a predicted risk label.

This AI component is strictly encapsulated as a stateless prediction service. It does not store patient data, trigger actions, or make autonomous decisions. All outputs are consumed by downstream system components and presented to clinicians as decision support signals.

6. Prototype Implementation

This section describes the implemented proof of concept prototype, demonstrating how the proposed AI system operates end to end within a controlled, locally deployed environment.

6.1 Scope of Implementation

The prototype implements the complete decision support pipeline, including: A frontend interface for clinicians to manage patients and sessions Backend services for authentication, session handling, and data storage A dedicated AI inference service for depression risk prediction

Certain aspects, such as large scale clinical integration and longitudinal deployment, are intentionally out of scope and deferred to future work.

6.2 AI Inference Service (FastAPI)

The AI model is deployed as a standalone FastAPI service. The service defines a structured input schema, applies the same preprocessing steps used during training, and returns model predictions in real time. This separation of concerns ensures: Clear system boundaries Independent testing and monitoring of the AI component Easier replacement or retraining of models in the future

The inference service operates locally to reduce privacy risks associated with transmitting sensitive mental health data.

6.3 Backend and Frontend Integration

The backend application orchestrates data flow between the frontend interface and the AI inference service. Clinician entered session text is securely transmitted to the prediction endpoint, and the resulting risk indicator is displayed within the session interface.

The frontend is designed to emphasize clarity and caution. AI outputs are framed explicitly as risk indicators, accompanied by contextual information and disclaimers reinforcing that clinical judgment remains paramount.

6.4 Local Deployment

All components of the prototype are deployed locally using a service based architecture. This design choice supports privacy preservation, reproducibility, and controlled experimentation, which are essential in healthcare oriented AI systems.

6.5 Backend Architecture (Node.js + MongoDB)

The backend layer is implemented using Node.js and serves as the central orchestration component of the system. Its primary responsibility is not machine learning inference, but the reliable and secure management of clinical workflows, user authentication, and persistent data storage.

A MongoDB database is used to maintain three core collections: psychologists, patients, and clinical sessions. This separation reflects real world clinical data organization and enables longitudinal tracking of patient history across multiple sessions. Persistent storage is a critical requirement in mental health contexts, where historical trends and prior assessments inform future clinical decisions.

Authentication and access control are enforced using JSON Web Tokens (JWT), ensuring that only authorized mental health professionals can access patient data and AI generated outputs. Importantly, the backend is deliberately decoupled from the AI inference service. This separation of concerns reduces system complexity, improves security, and allows the AI component to be independently tested, monitored, or replaced without affecting core application logic.

6.6 Frontend Interface (React based Human in the Loop UI)

The frontend interface is implemented using React and represents the primary point of interaction between human decision makers and the AI system. From an AI Systems Engineering perspective, the frontend plays a critical role in enforcing human agency and preventing automation bias.

The interface supports key clinician workflows, including secure authentication, patient management, session based text entry, and review of historical session outcomes. AI generated outputs are displayed within the session context as probabilistic risk indicators, clearly distinguished from diagnoses or recommendations. This design choice reinforces the decision support nature of the system and ensures that clinicians remain fully accountable for final judgments.

By embedding AI outputs within a structured, clinician controlled interface, the frontend operationalizes human in the loop principles and aligns with Trustworthy AI requirements related to transparency, oversight, and accountability.

7. Testing & Evaluation

Testing in this project goes beyond conventional model accuracy assessment and focuses on system level trustworthiness, as required in safety critical healthcare applications. The goal is to evaluate not only *what* the model predicts, but *how* the overall AI system behaves under uncertainty, bias, and operational constraints.

7.1 Model Level Evaluation

Baseline predictive performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1 score, in order to establish a minimum level of functional validity. These metrics provide an initial indication of the model's ability to distinguish between high risk and low risk textual inputs. However, they are treated as necessary but insufficient indicators of system reliability in a clinical context.

Model evaluation was conducted on held out validation data to reduce overfitting and to approximate performance on unseen inputs. The selected model demonstrated stable performance across multiple runs, supporting its use as a decision support component rather than a standalone diagnostic tool.

7.2 Safety Oriented System Testing

Given the potential harm associated with incorrect mental health assessments, safety was treated as a first class system property. The primary safety risks identified include false negatives (missed risk) and false positives (unnecessary distress or intervention).

To mitigate these risks, the system enforces the following safety mechanisms: AI outputs are explicitly framed as *risk indicators*, not diagnoses All AI generated outputs require mandatory human review The system does not trigger alerts, recommendations, or automated actions

These measures ensure that responsibility and accountability remain with qualified mental health professionals at all times.

7.3 Robustness and Input Sensitivity Testing

Robustness testing was conducted qualitatively by evaluating system behavior under degraded or ambiguous input conditions. Test cases included: Very short or incomplete textual inputs Neutral or emotionally ambiguous language Noisy text containing spelling errors or informal phrasing

Under these conditions, the system exhibits reduced confidence rather than forced classification. This behavior is intentionally preferred in safety critical systems, as it avoids false certainty and encourages clinician judgment when model confidence is low.

7.4 Fairness and Bias Considerations

Potential sources of bias were identified at both data and system levels, including: Linguistic variation across demographic and cultural groups Dataset bias stemming from DAIC WOZ and social media derived text Differences in emotional expression styles

Rather than attempting to fully eliminate bias through automated means, the system adopts a risk aware mitigation strategy, including: Human in the loop review of all outputs Conservative prediction thresholds Explicit acknowledgment of model limitations within the interface

This approach aligns with best practices for deploying AI in sensitive domains where biased automated decisions could cause harm.

7.5 Risk–Test–Mitigation Mapping

To provide a structured view of system level testing, Table 1 summarizes key risks, corresponding test strategies, observed behavior, and mitigation mechanisms.

Table 1: System Level Risk Assessment and Mitigation

Identified Risk	Test Strategy	Observed Behavior	Mitigation Mechanism
False negatives	Validation on borderline samples	Reduced confidence scores	Mandatory human review
False positives	Testing on neutral text	Conservative predictions	No automated actions
Input noise	Noisy / incomplete text	Confidence degradation	Clinician judgment required
Dataset bias	Cross dataset inspection	Performance variation	Human oversight, transparency
Automation bias	UI review and workflow testing	No autonomous decisions	Human in the loop design

7.6 Summary of Testing Outcomes

Overall, the testing results indicate that while the AI component provides useful probabilistic risk signals, the safety and reliability of the system emerge primarily from its architectural design and operational constraints, rather than from model performance alone. By prioritizing conservative behavior, transparency, and human oversight, the system demonstrates suitability as a clinical decision support tool rather than an automated diagnostic system.

8. Deployment & Reproducibility

This project emphasizes reproducibility and controlled deployment, which are critical requirements for AI systems operating in sensitive domains such as healthcare. All components are deployed locally to minimize privacy risks and to allow transparent inspection of system behavior.

8.1 Deployment Architecture

The system is deployed using a service oriented local architecture consisting of three main components:

- A FastAPI based AI inference service responsible for text preprocessing and prediction.
- A Node.js backend service responsible for authentication, session management, and database access.
- A React based frontend interface used by clinicians to interact with the system

The AI inference service loads a serialized TF IDF vectorizer and trained Naive Baiyes machine learning model at startup. This ensures consistent behavior across inference requests and avoids retraining during deployment.

8.2 Running the AI Inference Service

The AI component is exposed through a FastAPI application. At startup, the service loads the `tfidf.pkl` and `best_model.pkl` artifacts and initializes the preprocessing pipeline. The inference workflow consists of:

1. Receiving raw textual input via a POST request.
2. Applying preprocessing steps (lowercasing, stop word removal, stemming).
3. Transforming the text using the stored TF IDF model.
4. Generating a prediction using the trained Naive Baiyes model.
5. Returning the predicted risk label

This service is executed locally using a standard ASGI server, enabling real time inference while maintaining isolation from other system components.

8.3 Backend and Frontend Execution

The backend service is started independently and connects to a MongoDB instance to persist clinician, patient, and session data. Authentication tokens are generated at login and validated on each protected request, ensuring secure access to sensitive data.

The frontend application is launched separately and communicates exclusively with the backend and AI services through defined API endpoints. This separation enables independent development, testing, and replacement of system components.

8.4 Reproducibility Instructions

To reproduce the system locally, the following steps are required:

1. Start the MongoDB service.
2. Launch the Node.js backend application.
3. Start the FastAPI inference service.
4. Run the React frontend application.

All model artifacts, preprocessing logic, and inference code are version controlled. This setup allows other researchers or practitioners to reproduce results, inspect system behavior, and extend the system for future experimentation.

9. Discussion & Limitations

Key limitations include dataset bias, linguistic variation, and limited generalization across clinical contexts. These are mitigated through human oversight and conservative system behavior.

10. Conclusion & Future Work

This project demonstrates that AI can be responsibly integrated into mental health assessment when it is explicitly designed as a human in the loop decision support system, with carefully defined boundaries between automated analysis and human judgment. Future work includes expanded datasets, improved fairness evaluation, and controlled clinical studies.

References

1) *https://dcapswoz.ict.usc.edu/daic-woz-database-download/*
2) *https://www.kaggle.com/datasets/nikhileswarkomati/suicide-watch*

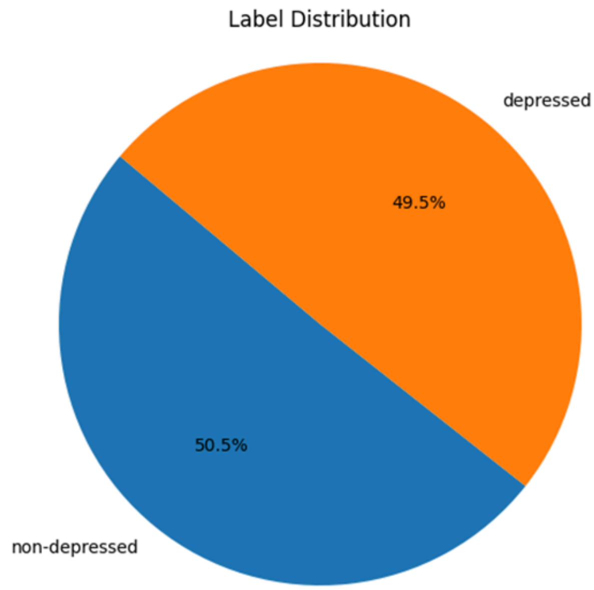
Appendix

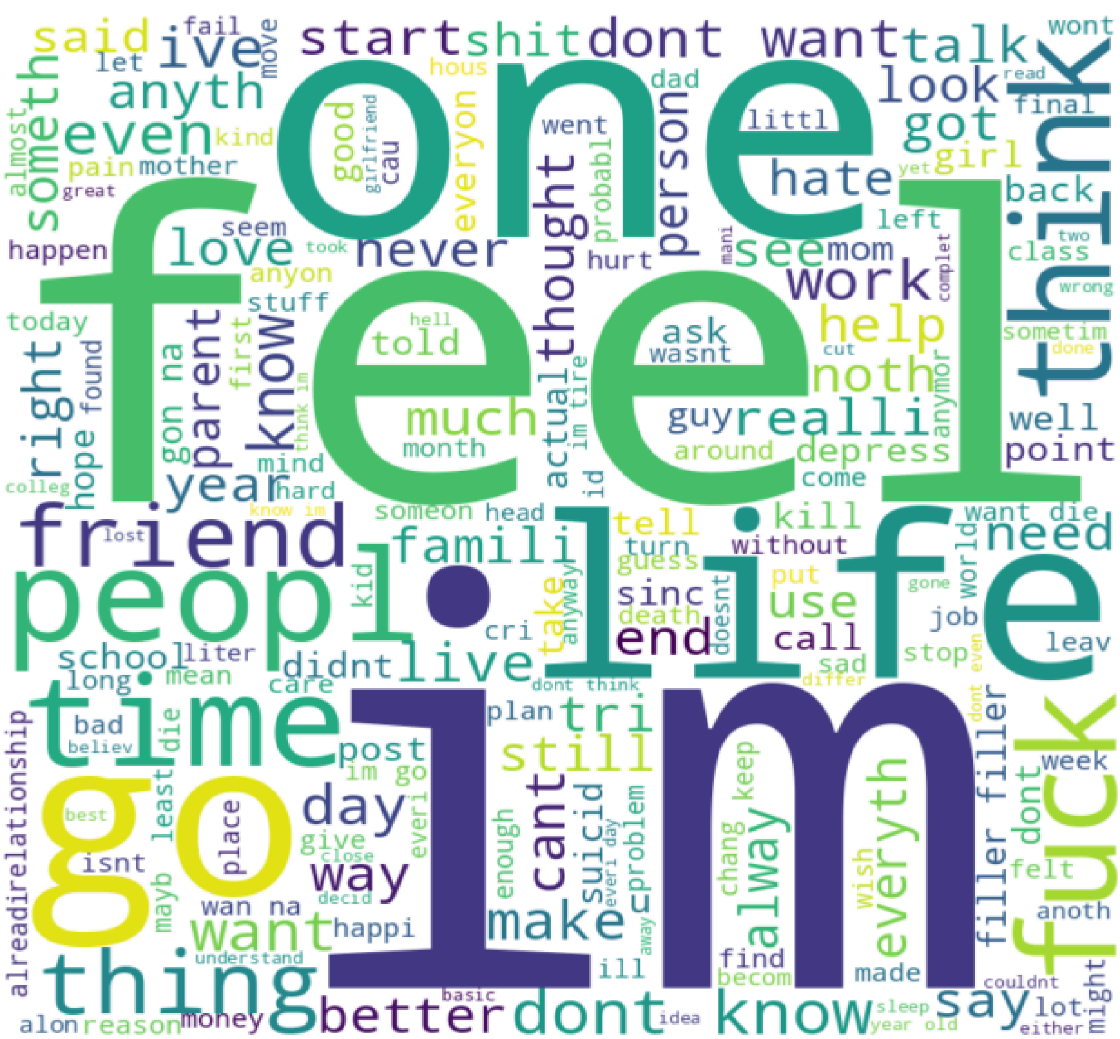
Detailed frontend, backend, preprocessing steps, and implementation specifics are provided here to support reproducibility without overloading the main report.

1. Snapshot of the dataset

161	I'm a misogynist Or just a misanthrope?	No depression
163	Advice for school kids Since I am the type to yell out wrong answer in Kahoot, a lot of people decides	No depression
164	Check out this new horror short I made https://youtu.be/S50McngM1ws it took me a while to make	No depression
165	I just played Bedwars I just played bedwars with total strangers over Discord and my legs were shive	No depression
167	I believe in rebirth... Yes I do because various people die at young age so what about their deeds wh	No depression
169	I hate my wife She is stinky poopoo I will not divorce	No depression
171	people love talking when theyâ€™re talking about something they love talking about people love talk	No depression
174	she either likes me or desperately wants me to know that she doesnâ€™t or iâ€™m probably overthi	No depression
176	I can't live with this thought, I feel sick. During last summer, I developed a sense of high anxiety	Depression
177	Not good enoughDepressed for like 10 years.	Depression
178	I want to cry, god fucking damnit I want to cry. Growing up being ridiculed for crying, being bullied	No depression
179	I'm convinced my math teacher hates me I've been struggling with math and it's tough for.me my tea	No depression
180	no rewards for resistensehi reddit. i am an anon. i do not know why i am postting this reaaly.	Depression
181	At this point I'm convinced relationships just don't exist Like, how can they??	No depression
182	I just feel like there is no point delaying the inevitableI've graduated from one of the best	Depression
183	it's so calming and fun ain't it just playing assassins creed 4 black flag and sailing in the sea with a sha	No depression
185	aight its settled im getting a stainless steel chain then i can wear my Dagaz rune on it which will be su	No depression
186	why does the dark sing why does the dark sing why does the dark singBreaking	Depression
187	It do be sad girl hour's :/ Idk y but I'm like super big sad it sucks smh	No depression
188	Yâ€™all are very swag except... To all the pervs in this you arenâ€™t swag and a personal fuck you.	No depression

2. Data Visualization Diagrams



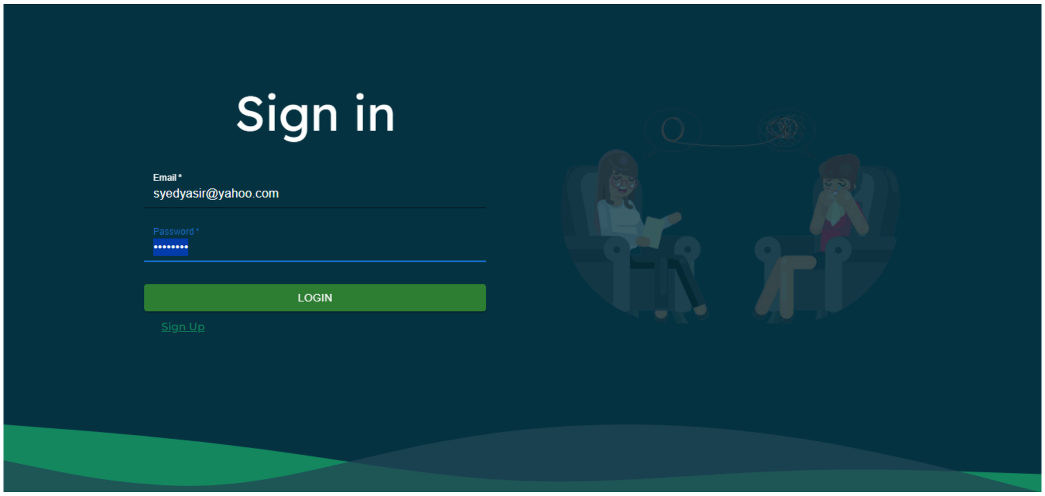


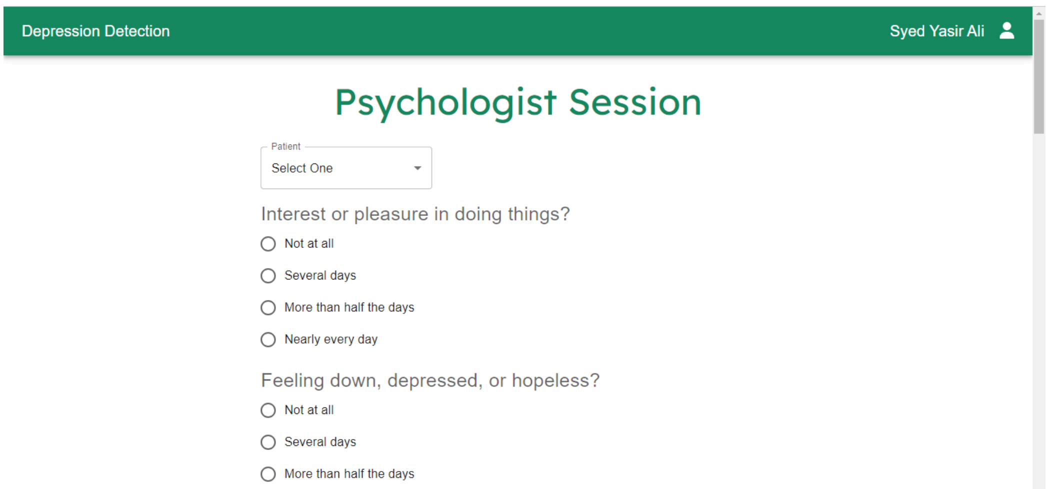
3. Naive Bayes Model Scores

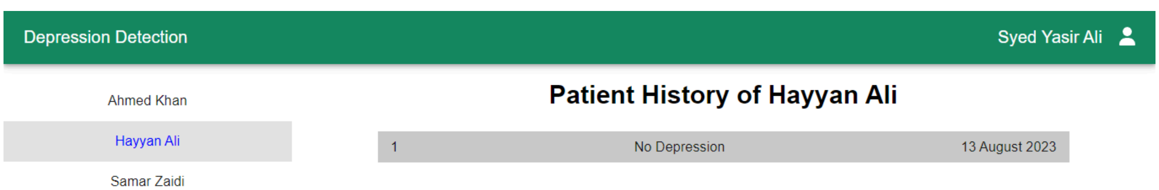
	precision	recall	f1-score	support
depressed	0.87	0.87	0.87	1458
non-depressed	0.88	0.88	0.88	1542

```
Training score: 0.899271324474925
Testing score: 0.8753333333333333
```

4. Front end React snapshots


Sign in
Email *
syedyasir@yahoo.com
Password *
LOGIN
Sign Up


Depression Detection
Syed Yasir Ali
Psychologist Session
Patient
Select One
Interest or pleasure in doing things?
Not at all
Several days
More than half the days
Nearly every day
Feeling down, depressed, or hopeless?
Not at all
Several days
More than half the days


Depression Detection
Syed Yasir Ali
Ahmed Khan
Hayyan Ali
Samar Zaidi
Patient History of Hayyan Ali
1
No Depression
13 August 2023